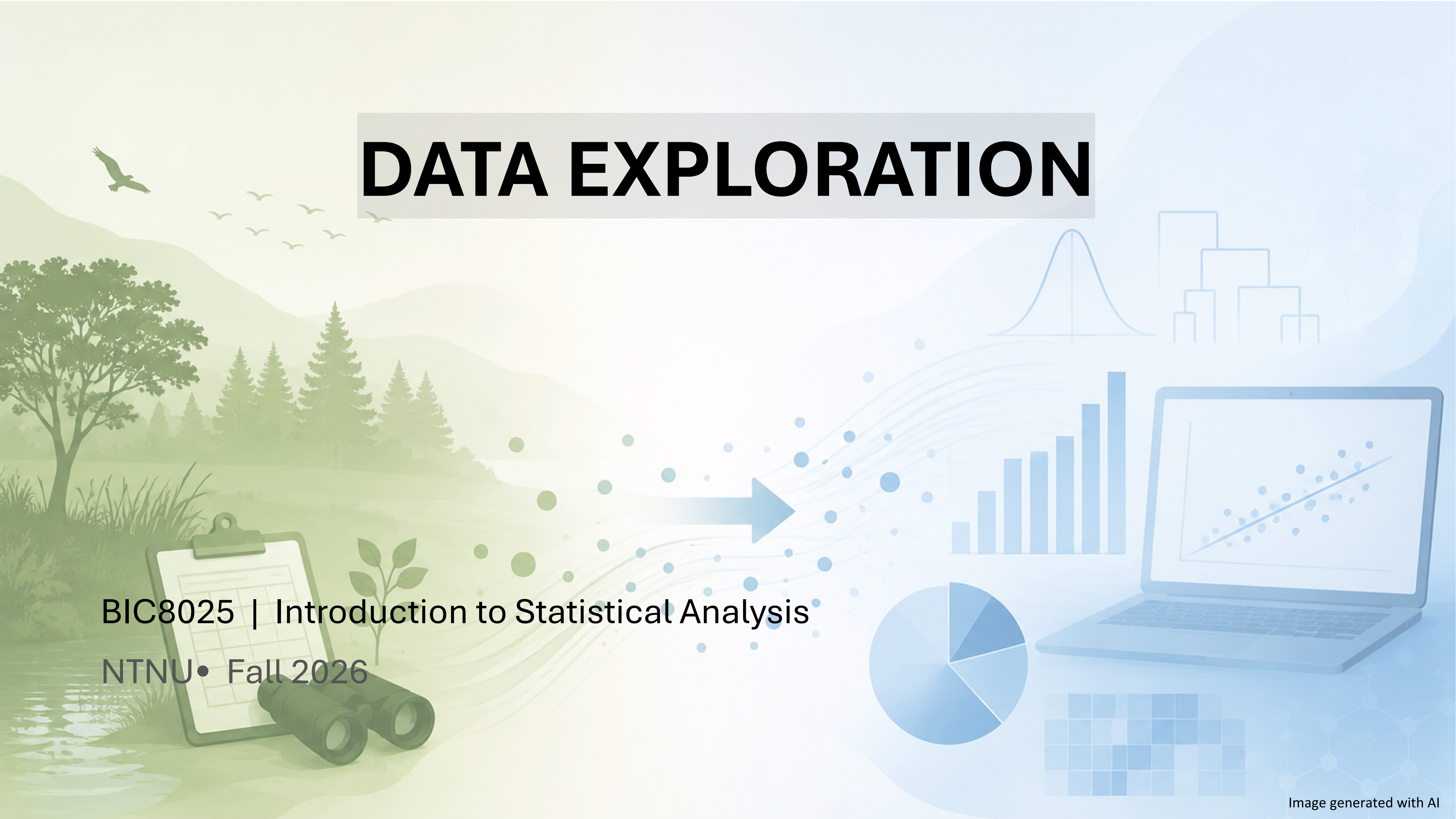
DATA EXPLORATION



BIC8025 | Introduction to Statistical Analysis

NTNU• Fall 2026

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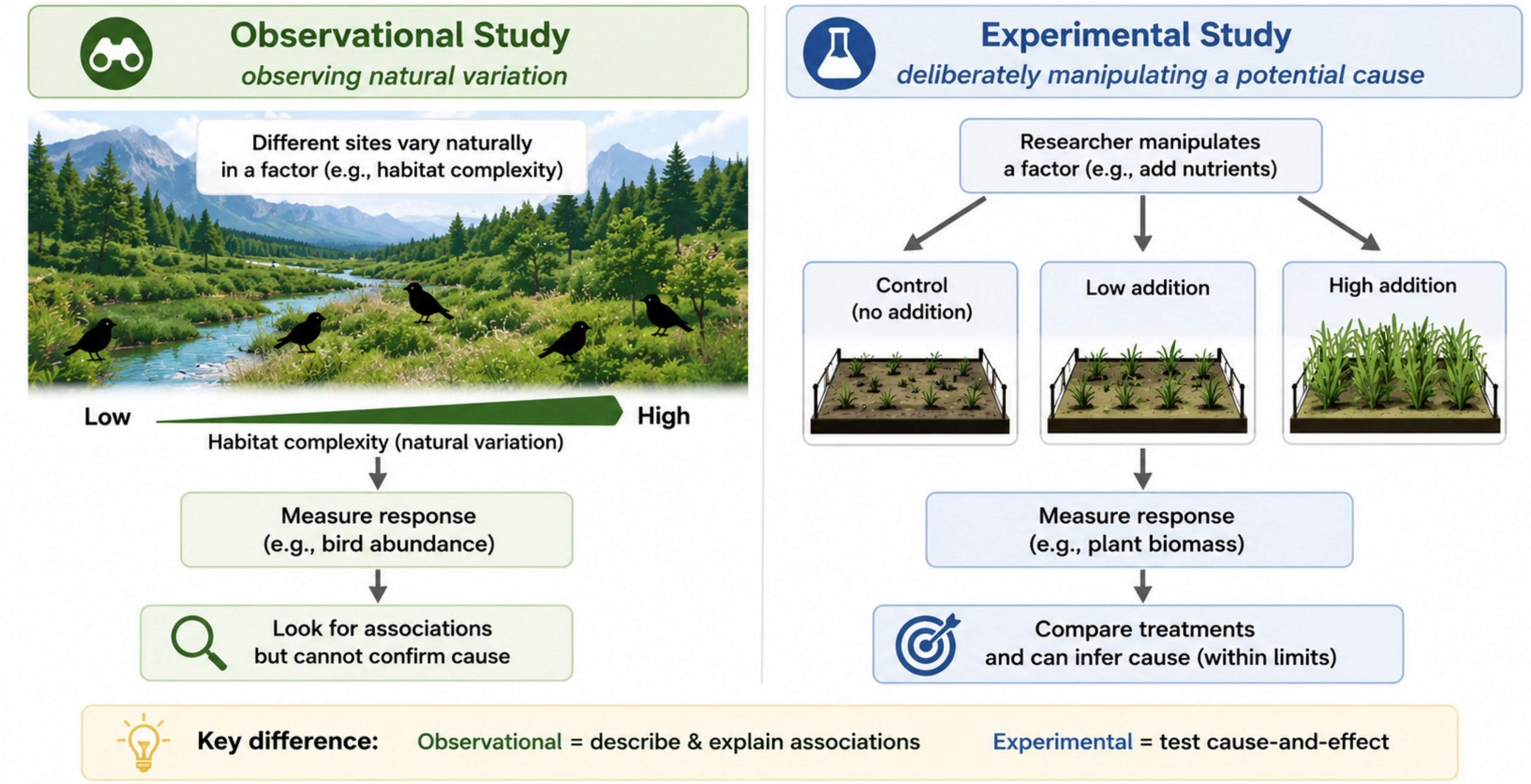
Two ways we get data



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Observational data

- The researcher measures variables as they naturally occur (sometimes over environmental gradients or from different space and time).
- Examples?

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Strength

- realistic field conditions
- broad spatial/temporal coverage
- useful when manipulation is impossible or unethical

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Issues

- association does not necessarily imply causation
- confounding factors
- spatial/temporal dependence
- non-random sampling

Experimental data

- The researcher manipulates explanatory variables (always creating artificial environmental gradients or groupings) and measure response variables
- Examples?

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Strength

- random assignment helps separate treatment effects from natural variation among experimental units
- stronger evidence for causal effects

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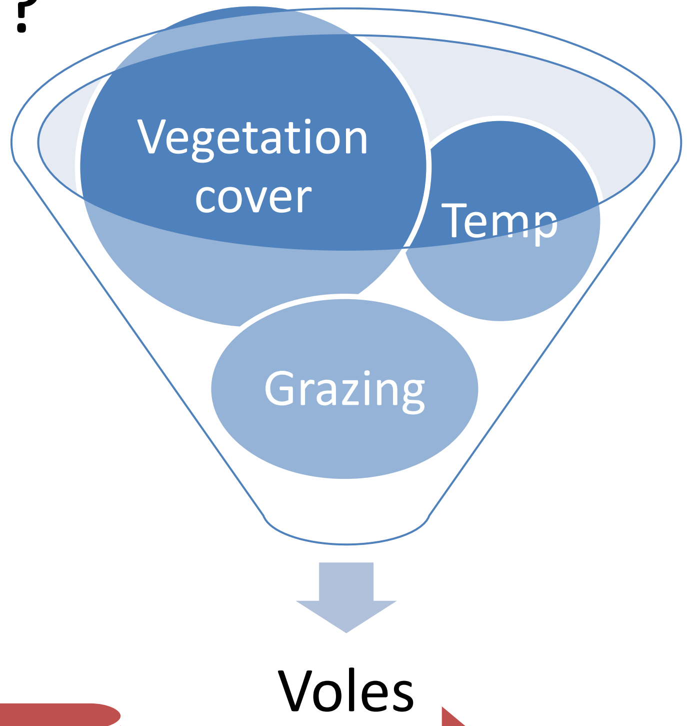
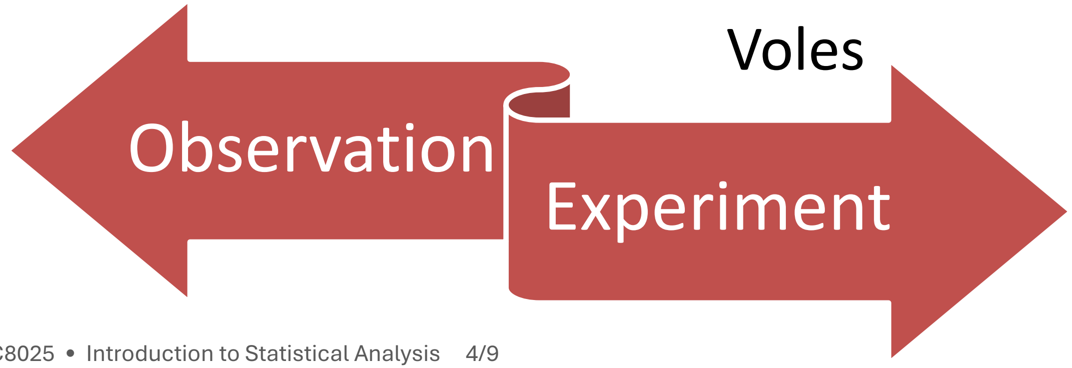
Issue

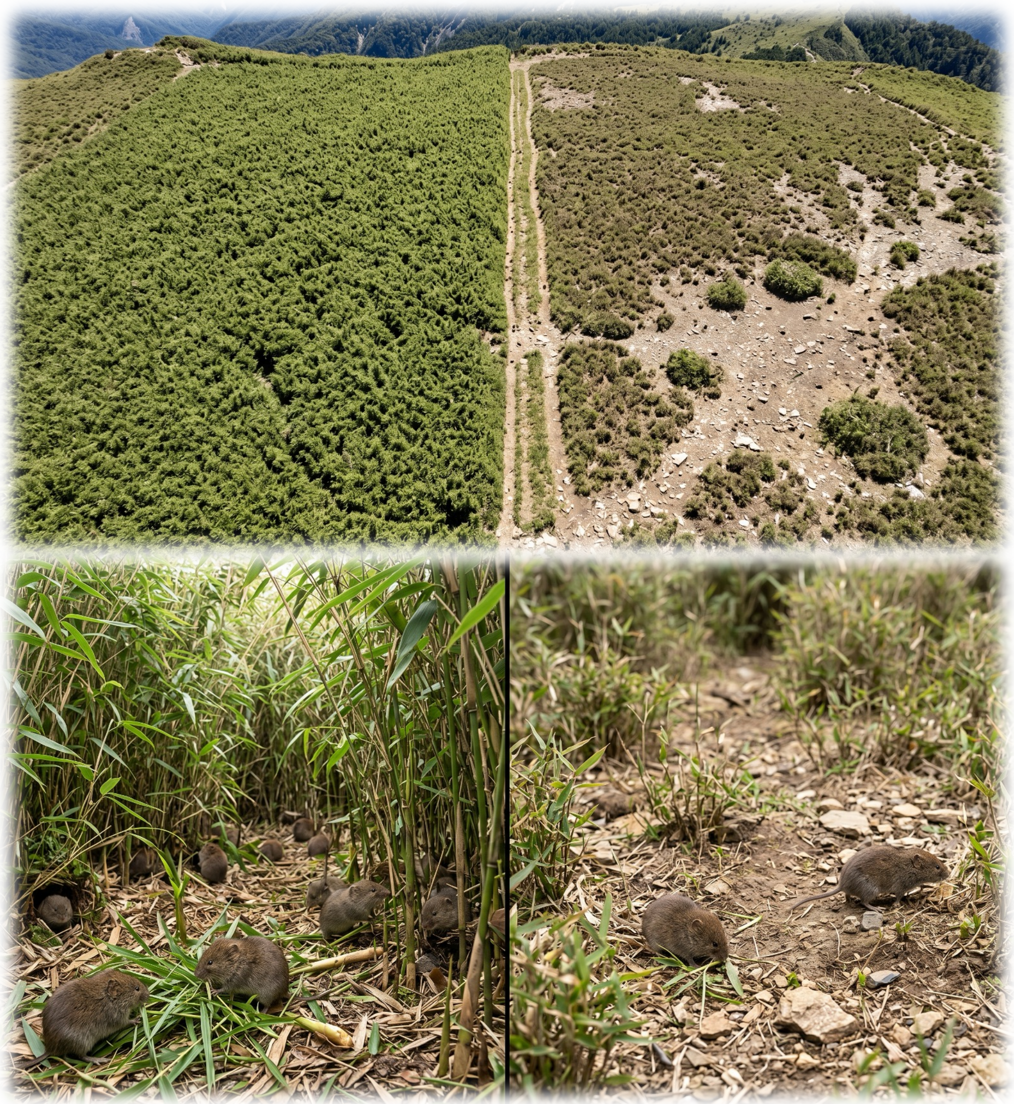
- unrealistic conditions if not carefully designed
- manipulation is more restricted in spatial/temporal coverage

Same question, experiments or observations?

- Factors influencing vole abundance?







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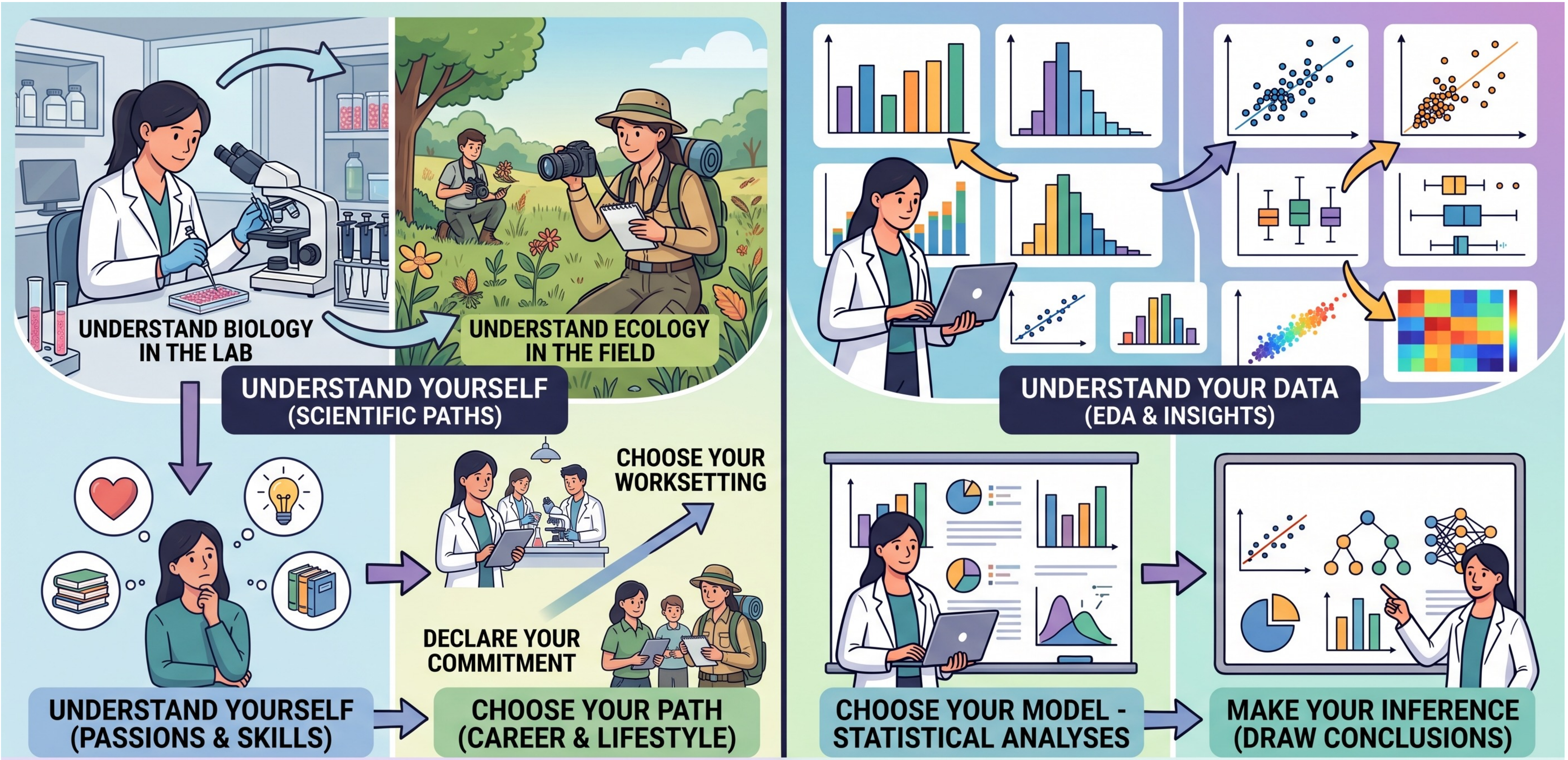
Exploratory data analysis (EDA)

- Statistical models make assumptions about data-collection process (e.g., random sampling, sample independence)
- Biological datasets, even with carefully-designed data-collection process, still commonly contain **outliers, non-normal distribution**, **unequal variance, zero inflation**, **collinearity**, **nonlinear relationships**, and **non-independence** due to underlying biological process.
- Be a disciplinary expert first, a statistician second (*or, help the statistician to help you by knowing your data well*)

Data (*Who You Are*) before model (*What You Do*)



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Exploratory data analysis: 8 steps

- Outliers
- Homogeneity
- Normality
- Non-linearity
- Collinearity
- Zero inflation
- Interaction
- Independence

The following illustrations are based on a *X*-*Y* relationship model: $Y \sim X$

Outliers

- **Why:** signs of measurement errors, indication of non-Gaussian error distribution or mixture of two distributions
- **What:** response Y, predictor X, residual (error)
- **How**: boxplots, Q-Q plots

Do not delete observations simply because they are extreme; first ask whether they are telling you the actual data distribution that could reflect biologically meaningful variation.

Outlier detection with the IQR method

- Statistical outliers: observations that lay outside the range of {Q1-1.5xIQR, Q3+1.5xIQR}
- Designed for: Gaussian models (normally distributed errors)
- My advice: use it as the last resort when you cannot figure out why there are extreme values

Outliers are opportunities to improve experimental or sampling protocols, guide statistical modeling, and generate new research questions.

Homogeneity of Variance (Homoscedasticity)

- **Why**: potential for biased residual (error) estimates and skewed inferences, indication of mis-specified model structures
- **What**: whether variance of residuals (errors) remains constant across all values or levels of predictor X.
- **How**: boxplots, Q-Q plots, residuals-vs-fitted plots

The question is not "*is the variance identical*" (it never is). It is: “*does the variance changes SYSTEMATICALLY*”?

Money doesn't buy happiness—

It buys the freedom to be miserable

(for entirely non-financial reasons)

Climate change ecology—

How important it is to consider heteroscedasticity in forecasting ecological consequences of climate change?

Normality

- **Why**: Gaussian models vs non-Gaussian models (e.g., generalized linear models such as Poisson or negative binomial models, Bayesian models, non-parametric models such as randomization models)
- **What**: residual (error)
- **How**: boxplots, Q-Q plots, residuals-vs-fitted plots

It is NOT the response Y or predictor X that needs to checked for normal distribution; is the residual (error).

Zero inflation

- **Why**: signs of measurement limitation or sampling bias (sampling zeros), or mixture of two distinct biological processes (structural zeros/true zeros, often indicating a different process from positive counts)
- **What**: response Y has zeros more often than expected
- **How**: histograms, boxplots

Do not jump in a zero-inflated model simply because there are excessive zeros; first ask whether the zeros are measurement limitation or sampling bias.

R Exercise: Outliers, Homogeneity, Normality, Zero Infection

- 2-3 students working in a group
- Install R (R studio optional)
- Download the data set (voles_raw.csv)
- Run the r script (EDA_exercise_steps1-4.r)
- Discuss the output
- Share the group findings

Ready to come out and explore?

Dataset

Dataset (voles_raw.csv)

- Live-trapping of grassland voles. 30 sites in 3 valleys (“valley”); each site trapped on 4 occasions (“month”, including April, June, August, October). Sampling effort (“trap_nights”) varies between sites and months.
- Response Y: number of voles caught (variable “voles”).
- Predictors: grazing (binary variable “grazing”), temperature (continuous variable “temp_c”) and vegetation cover (continuous variable “veg_cover_pct”)

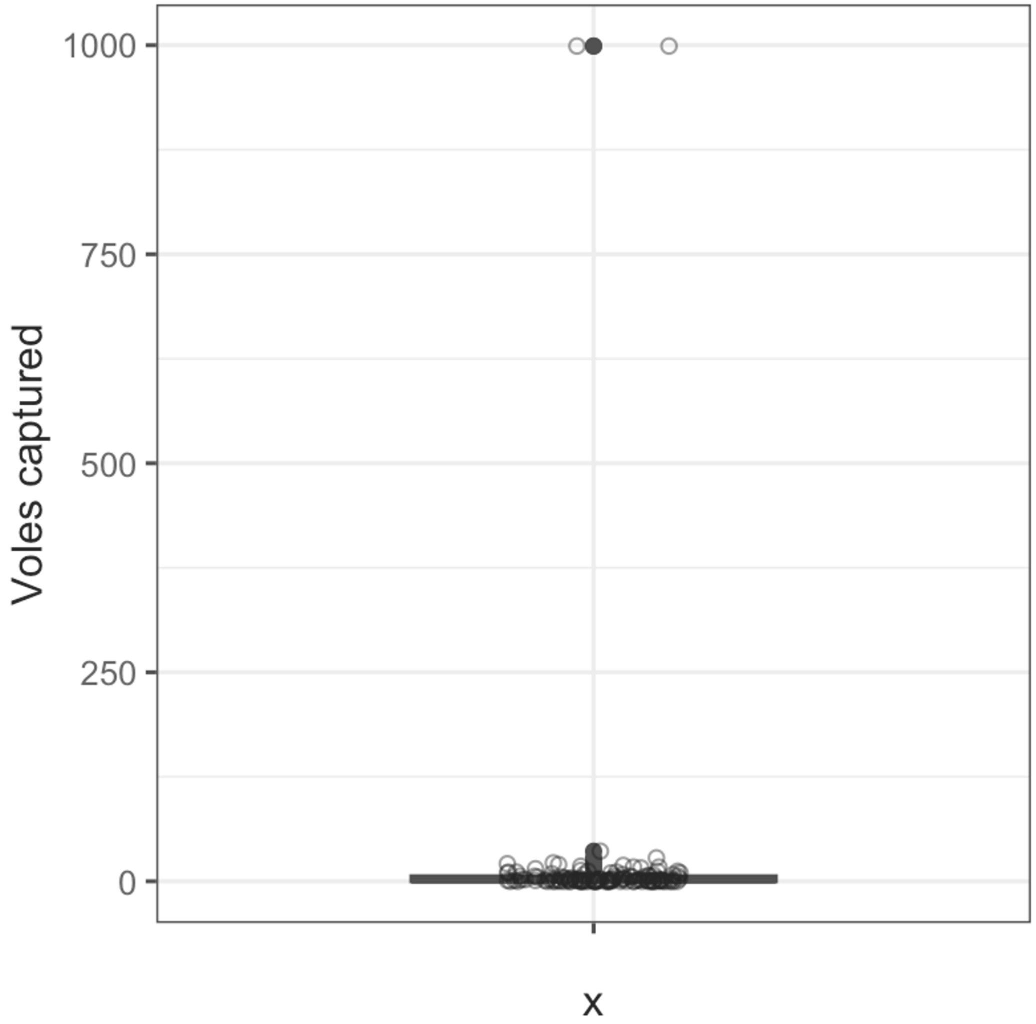
Research question

$vole\ abundance \sim vegetation\ cover + grazing?$

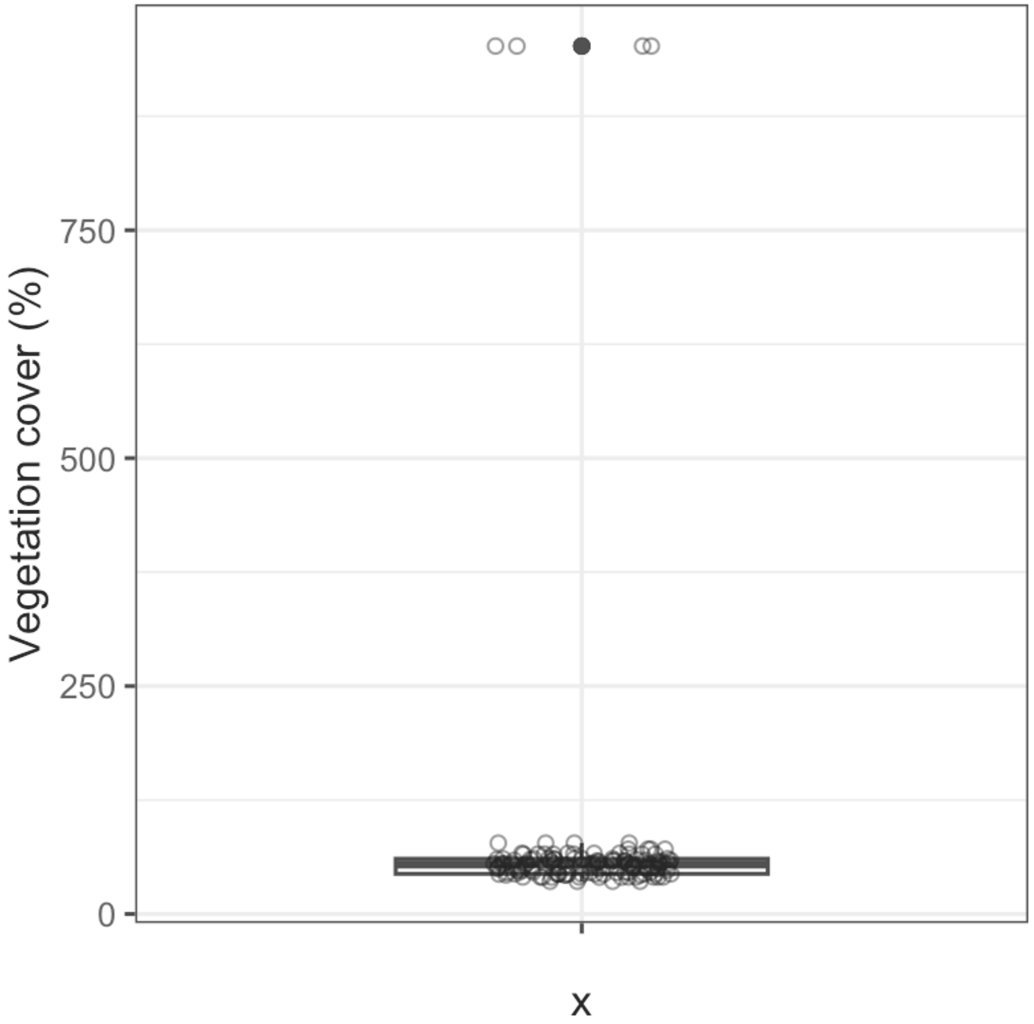
Background image generated with AI

Outliers?

Response (raw)

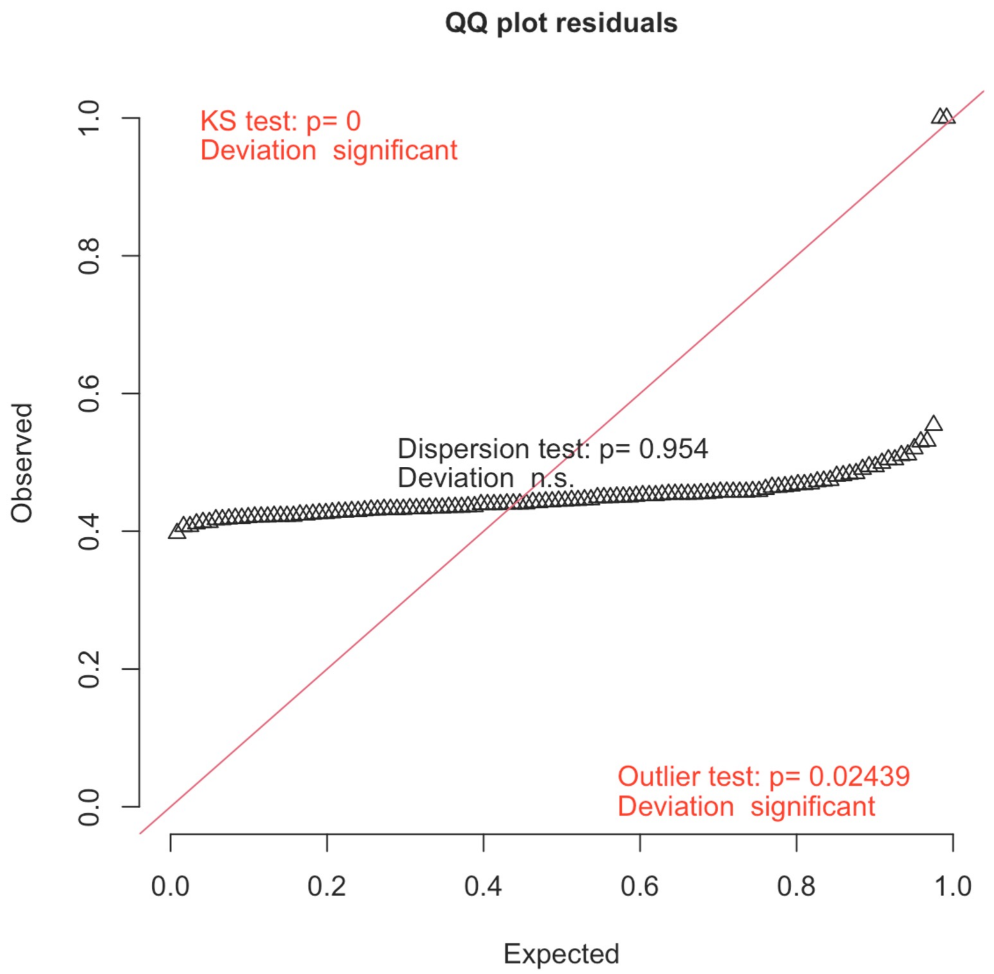


Predictor (raw)



Outliers?

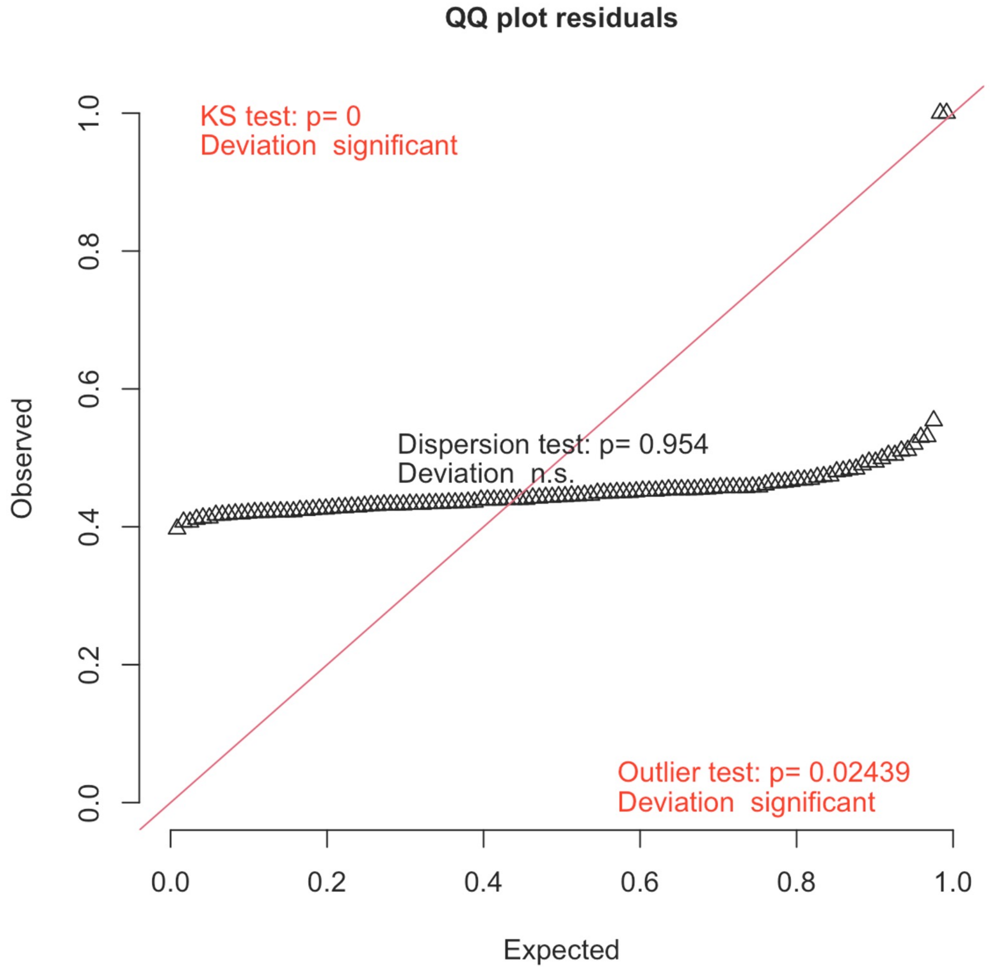
Residuals from naïve linear model



"Naïve" model:
Ignoring data exploration and direcrly jumping in statistical models that we are mostly familiar with (e.g., t-test, ANOVA, linear regression, linear correlation)

Q-Q plots

Residuals from naïve linear model



A Quantile-Quantile (Q-Q) plot is a scatter plot used to assess whether a dataset has unusual values, or follows a specific theoretical distribution (most commonly a normal distribution).

Homogeneity of variance?

$$Dispersion\ index = \frac{Variance}{Mean}$$

$$Dispersion = 1 \rightarrow Random\ count$$

$$Dispersion > 1 \rightarrow Overdispersion$$

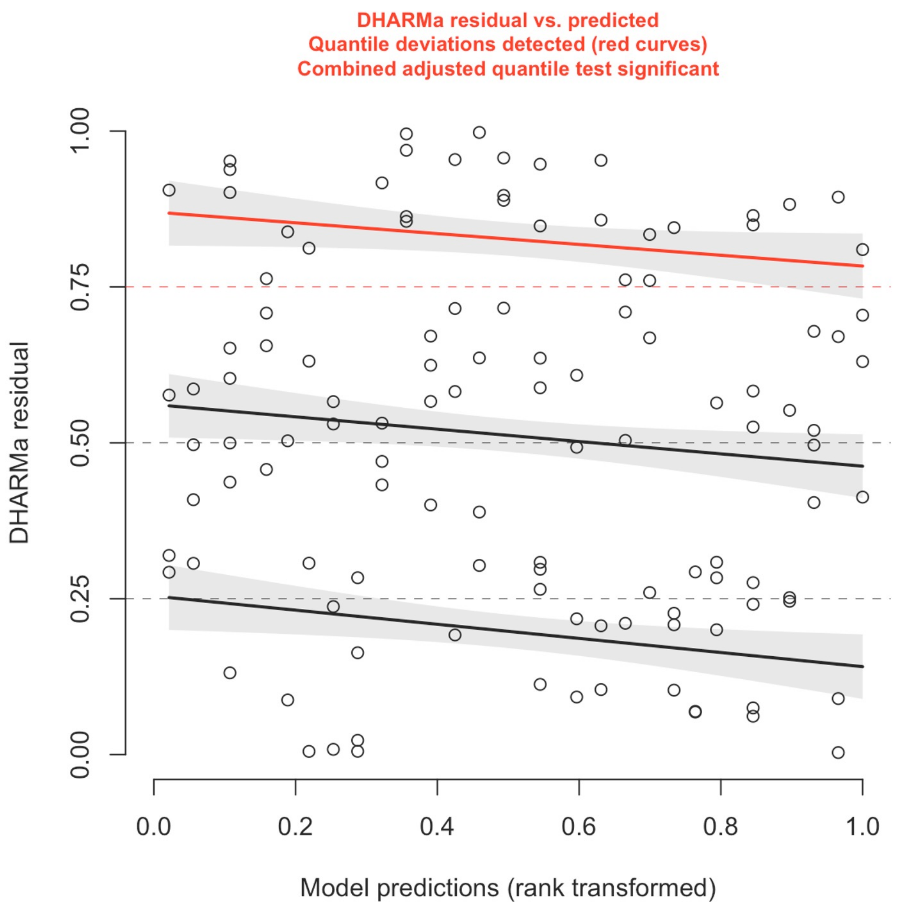
$$Dispersion < 1 \rightarrow Underdispersion$$

veg_cover_level	n	mean	variance	var_mean_ratio
<fct>	<int>	<dbl>	<dbl>	<dbl>
[35.9,44]	27	3	11.1	3.69
(44,50.9]	20	4.55	51.3	11.3
(50.9,55.9]	24	6.25	79.1	12.7
(55.9,63.7]	23	3.39	29.1	8.57
(63.7,95.2]	24	5.25	46.3	8.82

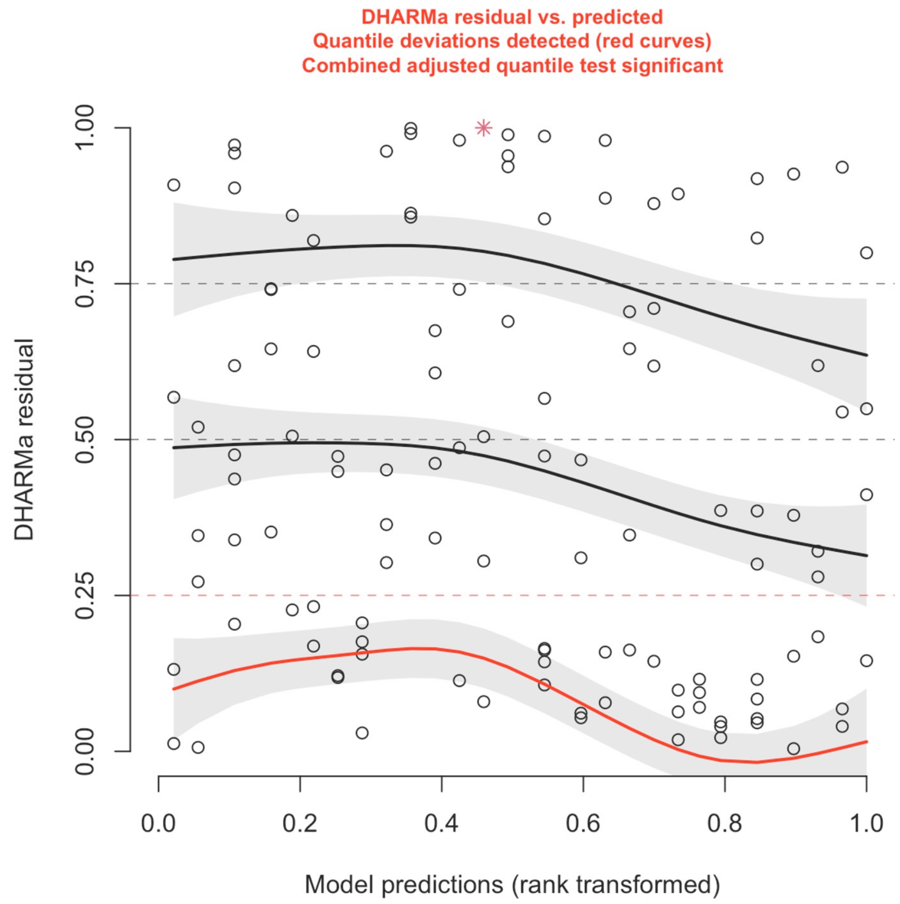
Homogeneity of variance?

Residuals vs predicted plots from a (not so naïve) negative binomial model

Heteroscedasticity



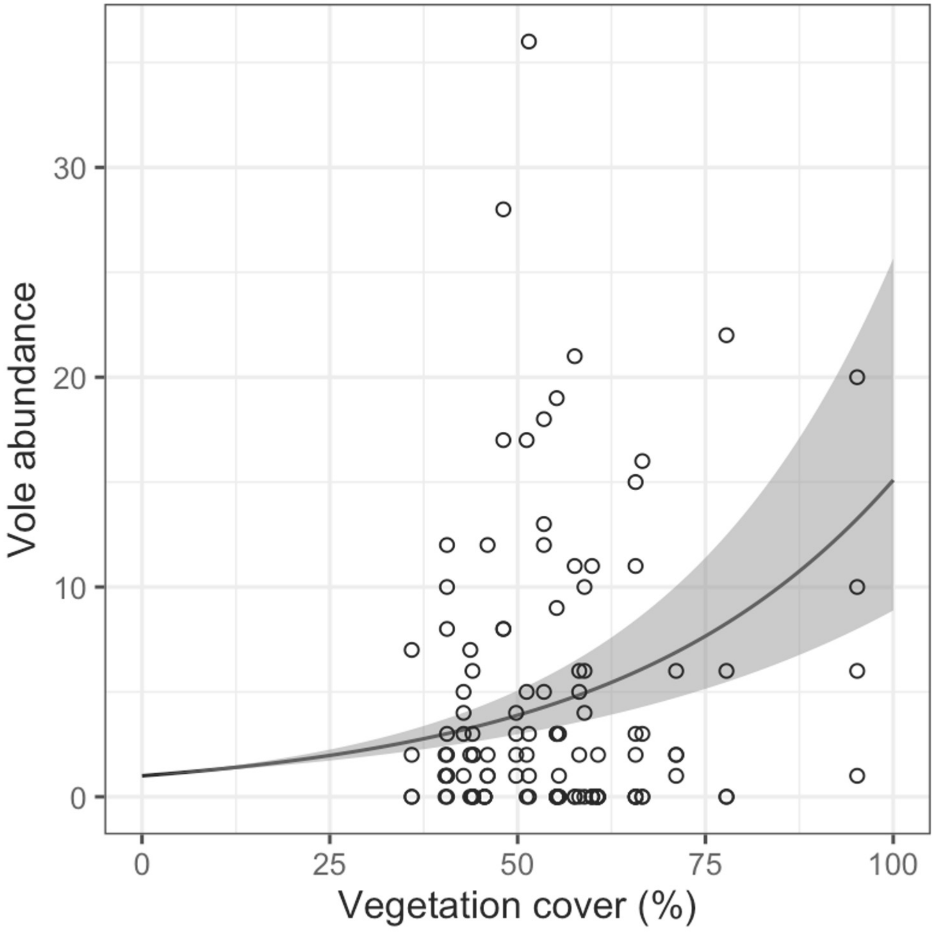
Homoscedasticity



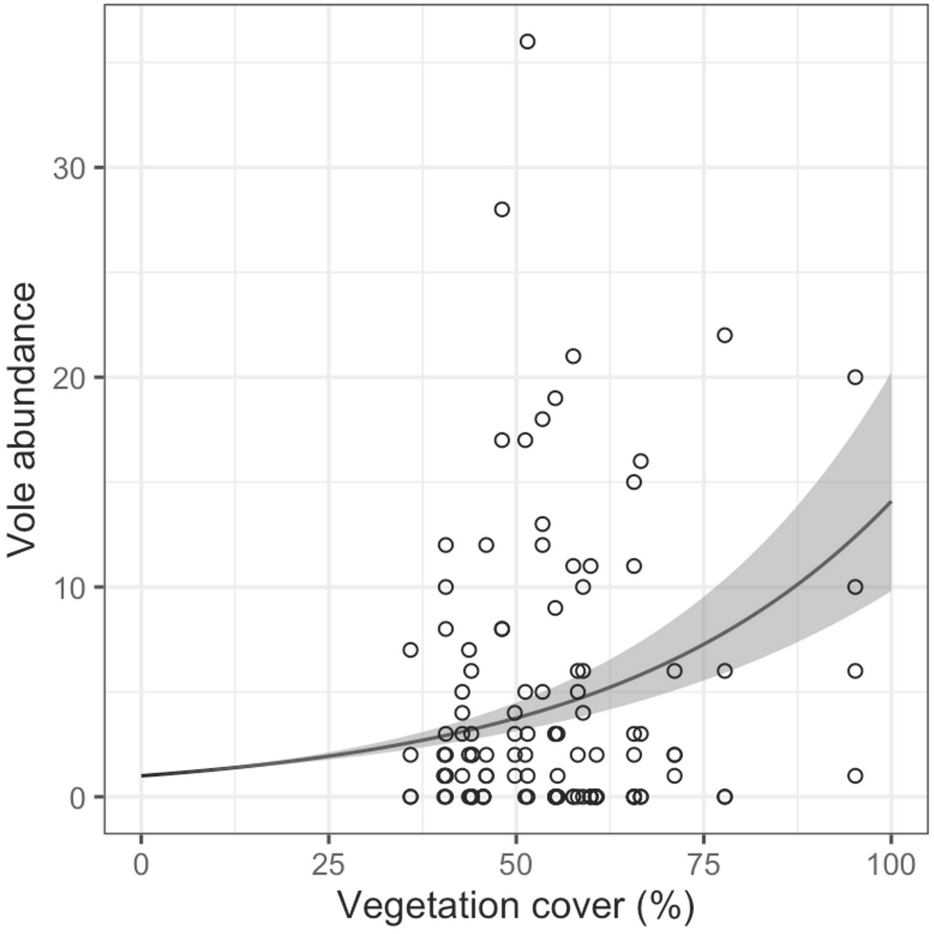
Homogeneity of variance?

Predicted response from a (not so naïve) negative binomial model

Heteroscedasticity

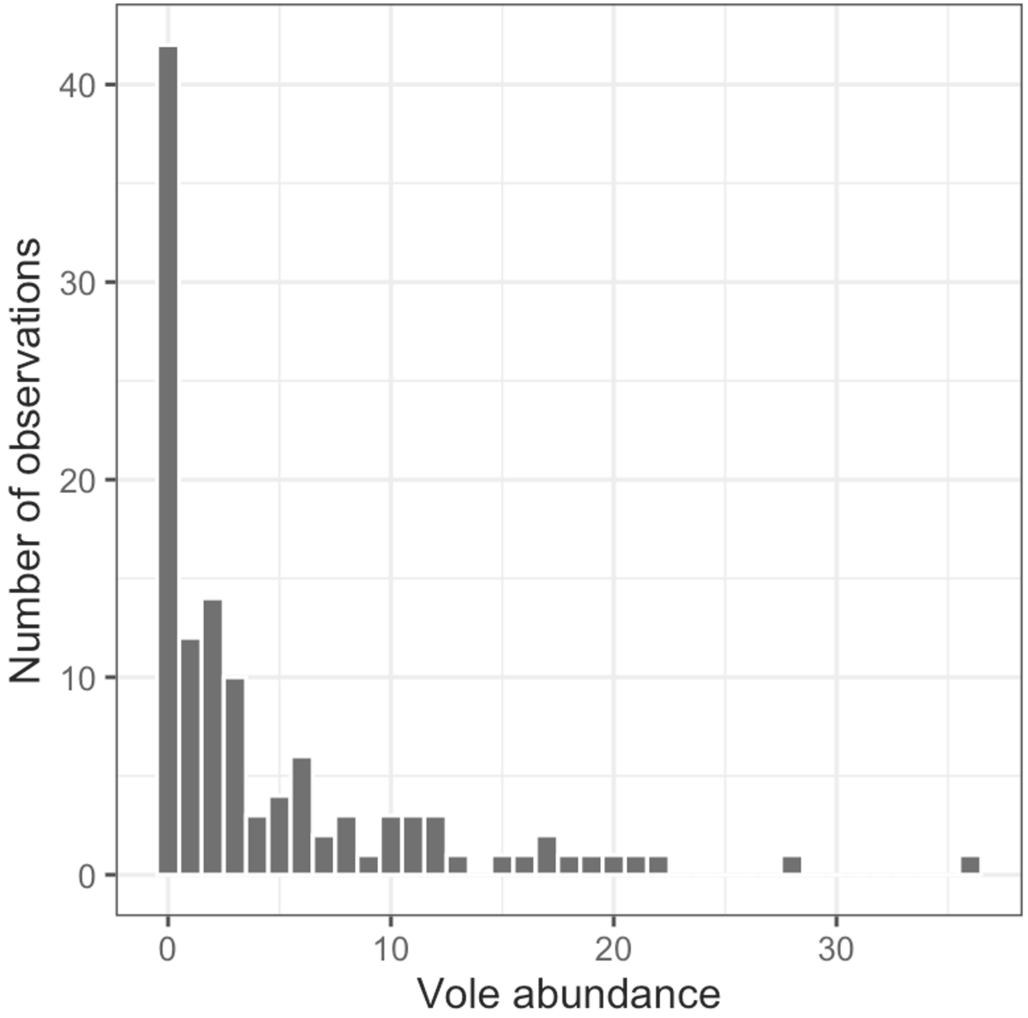


Homoscedasticity

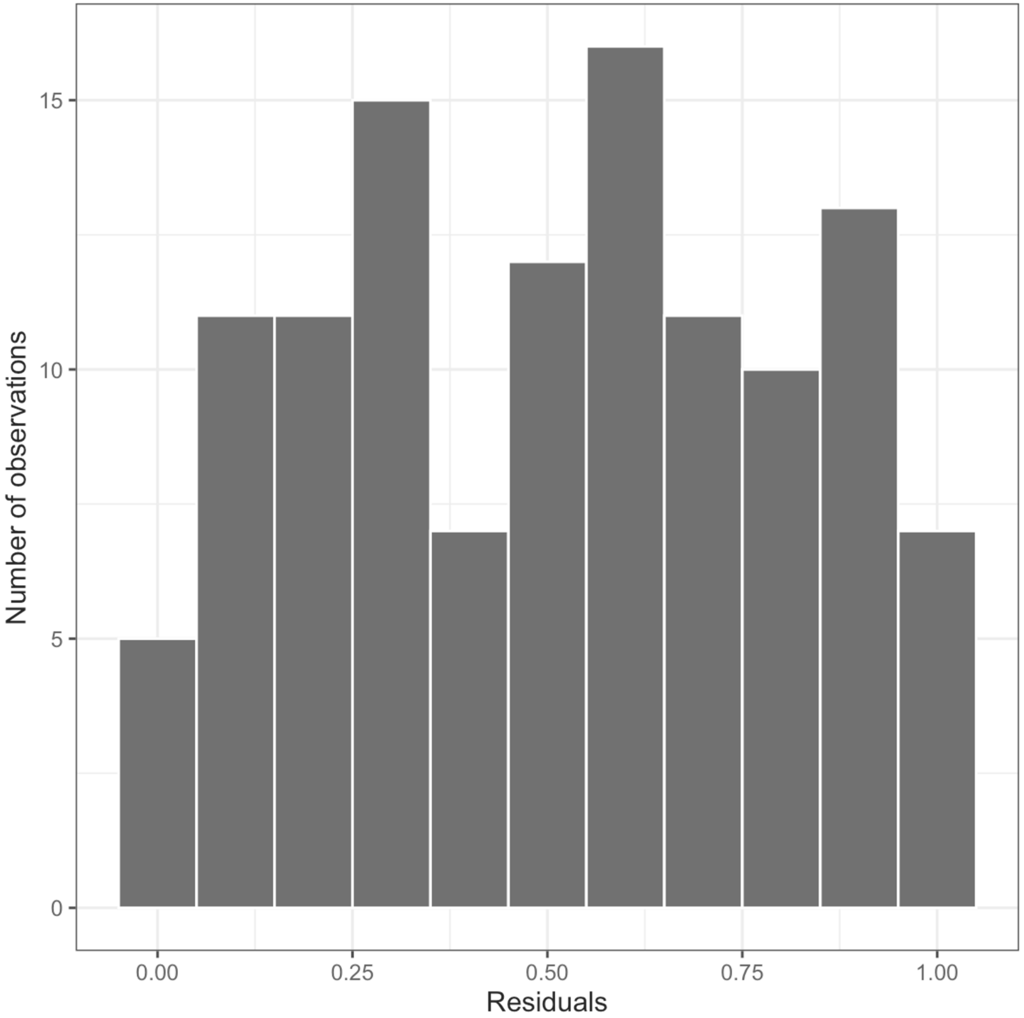


Normality?

Response (raw)

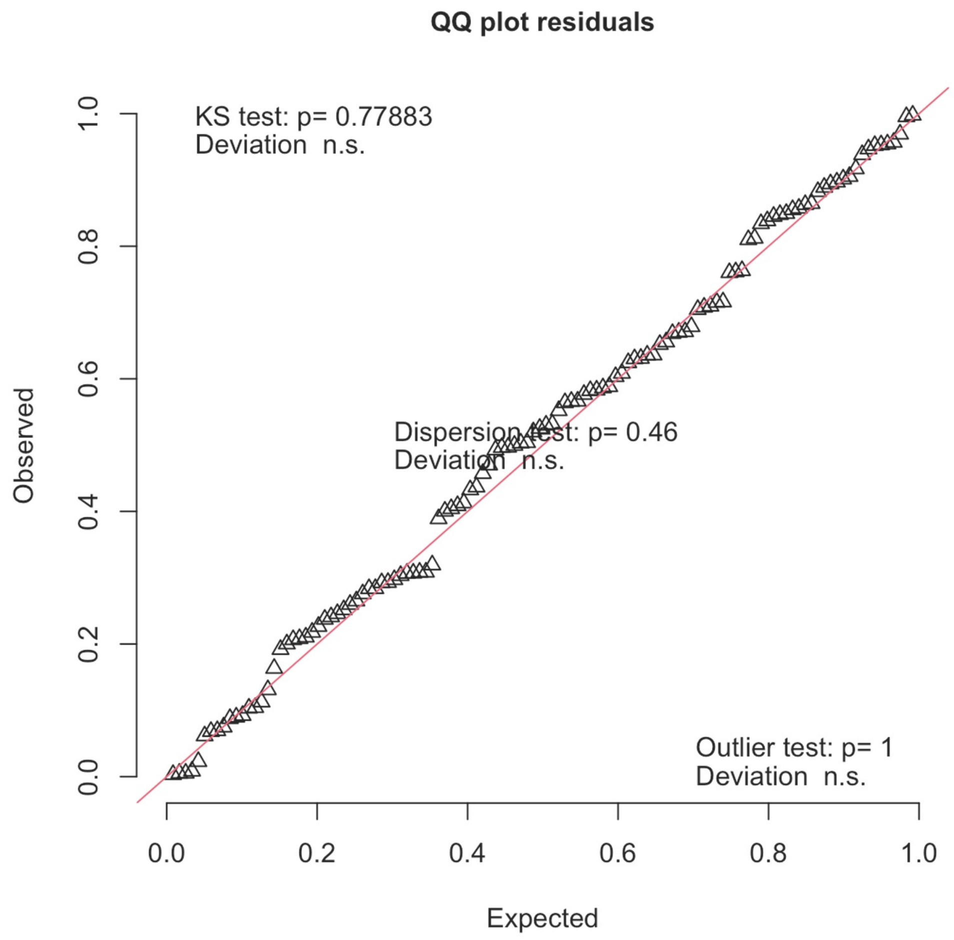


Residuals (Predicted – Raw)

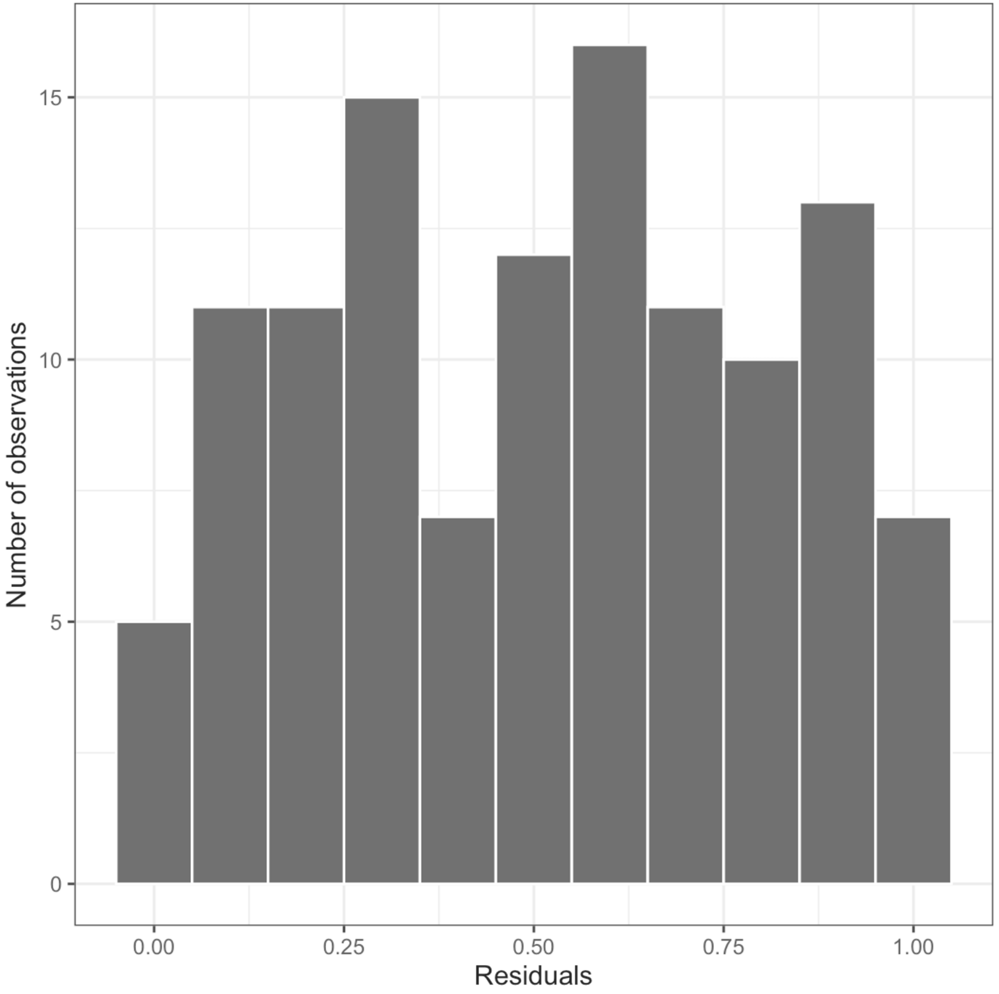


Normality?

Residuals (Predicted – Raw)

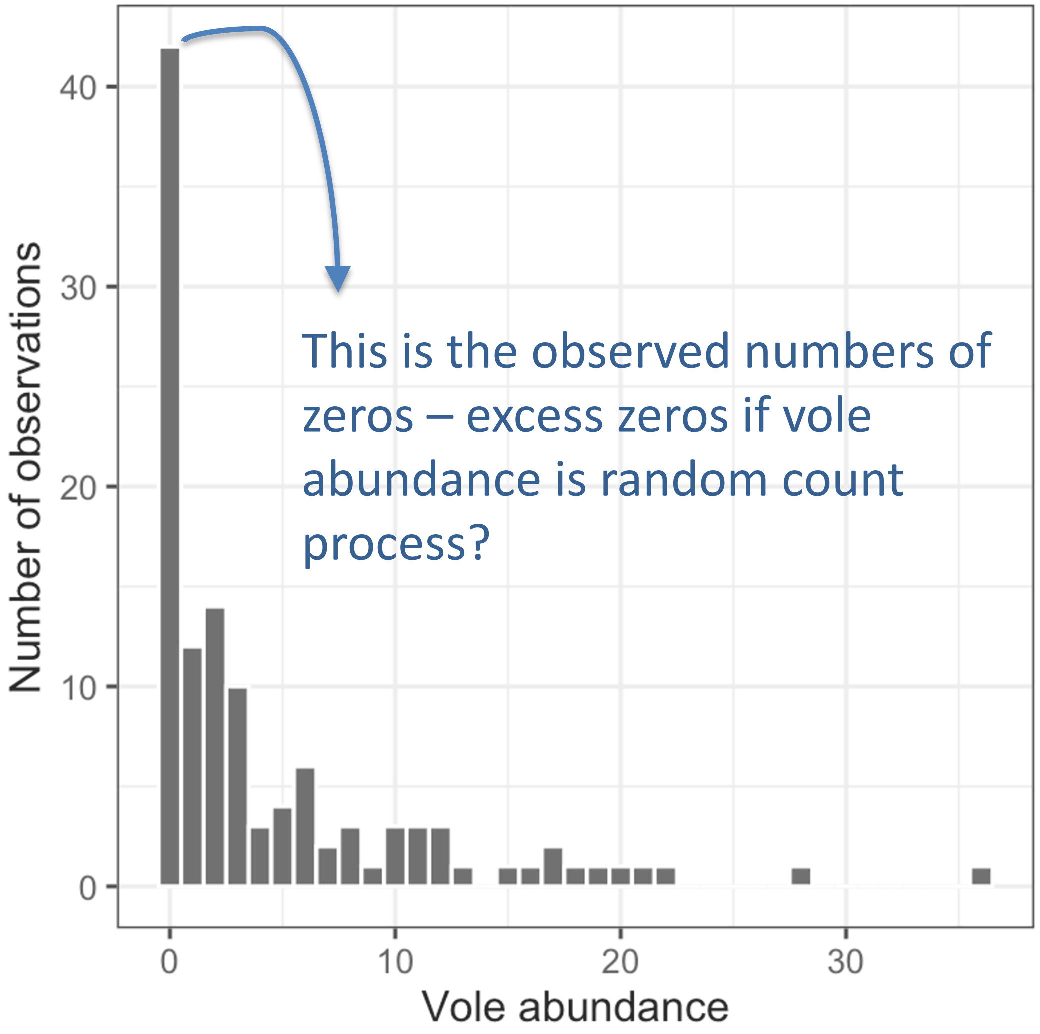


Residuals (Predicted – Raw)

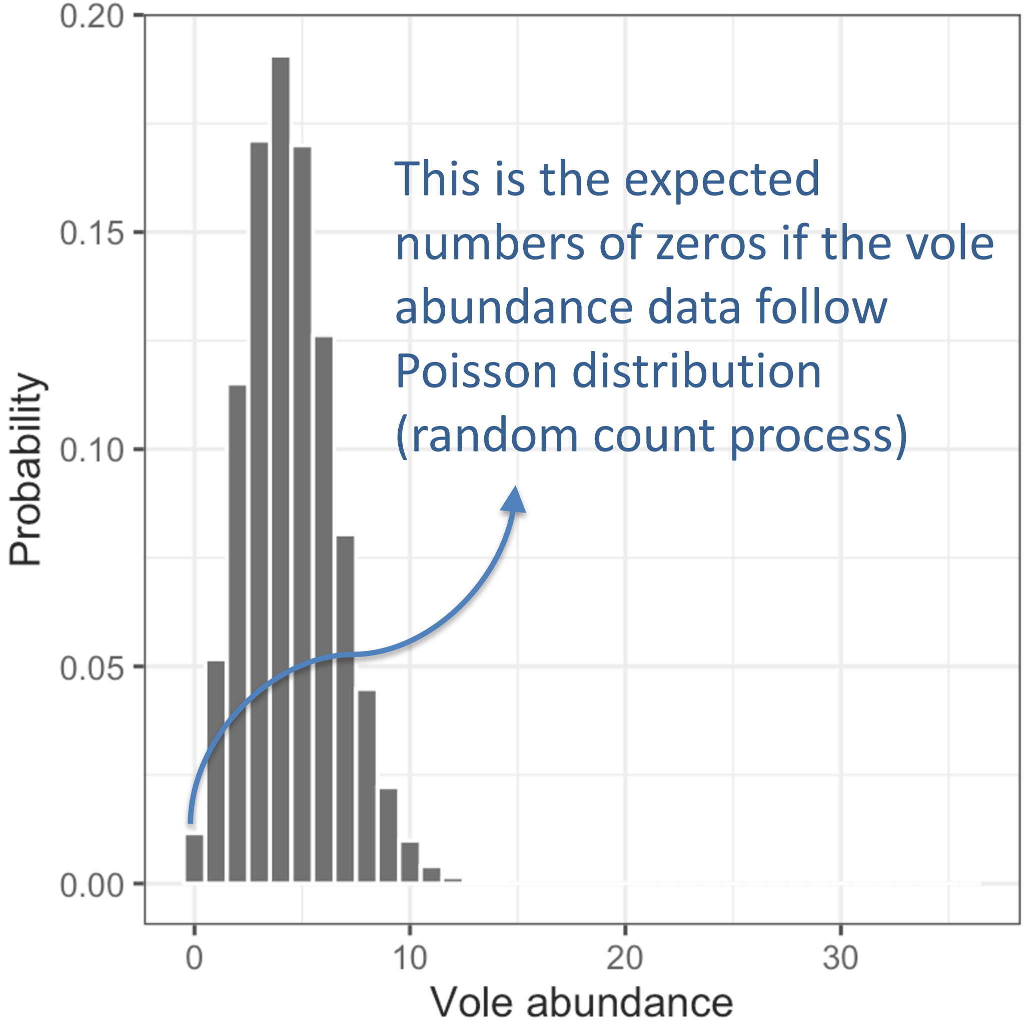


Zero inflation?

Response (raw)

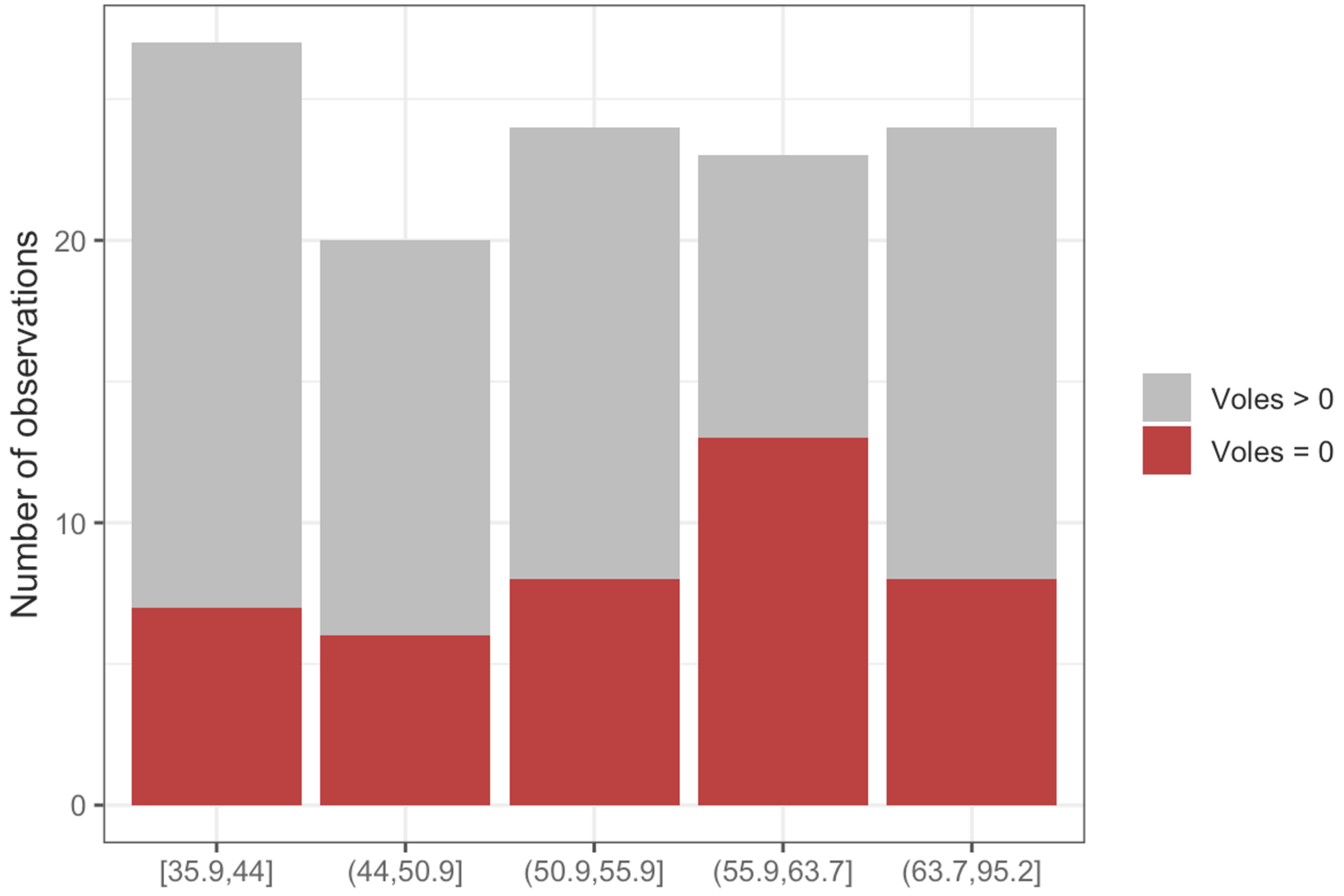


Response (theoretical Poisson)



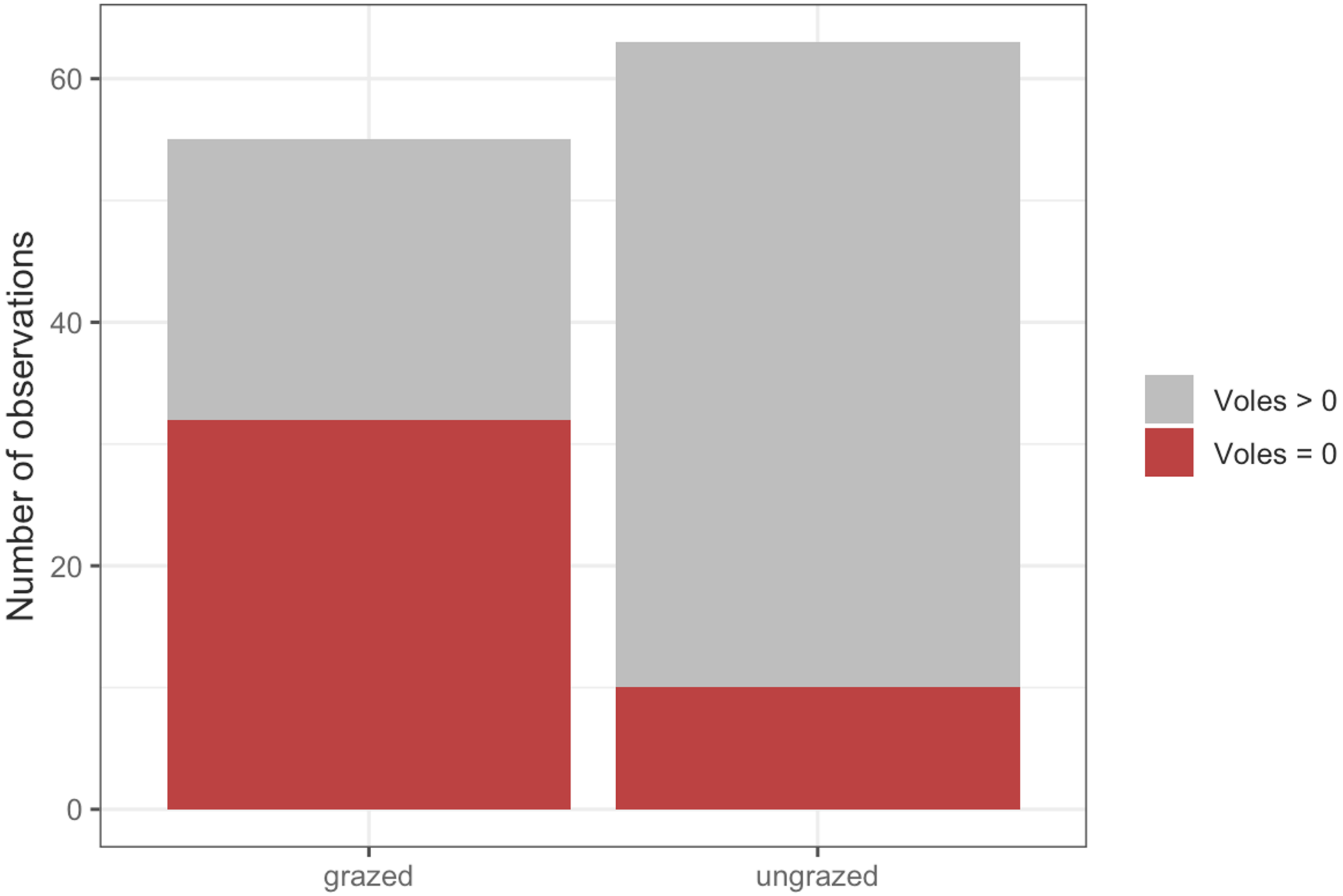
Zero inflation?

How are excess zeros distributed across levels of vegetation cover?



Zero inflation?

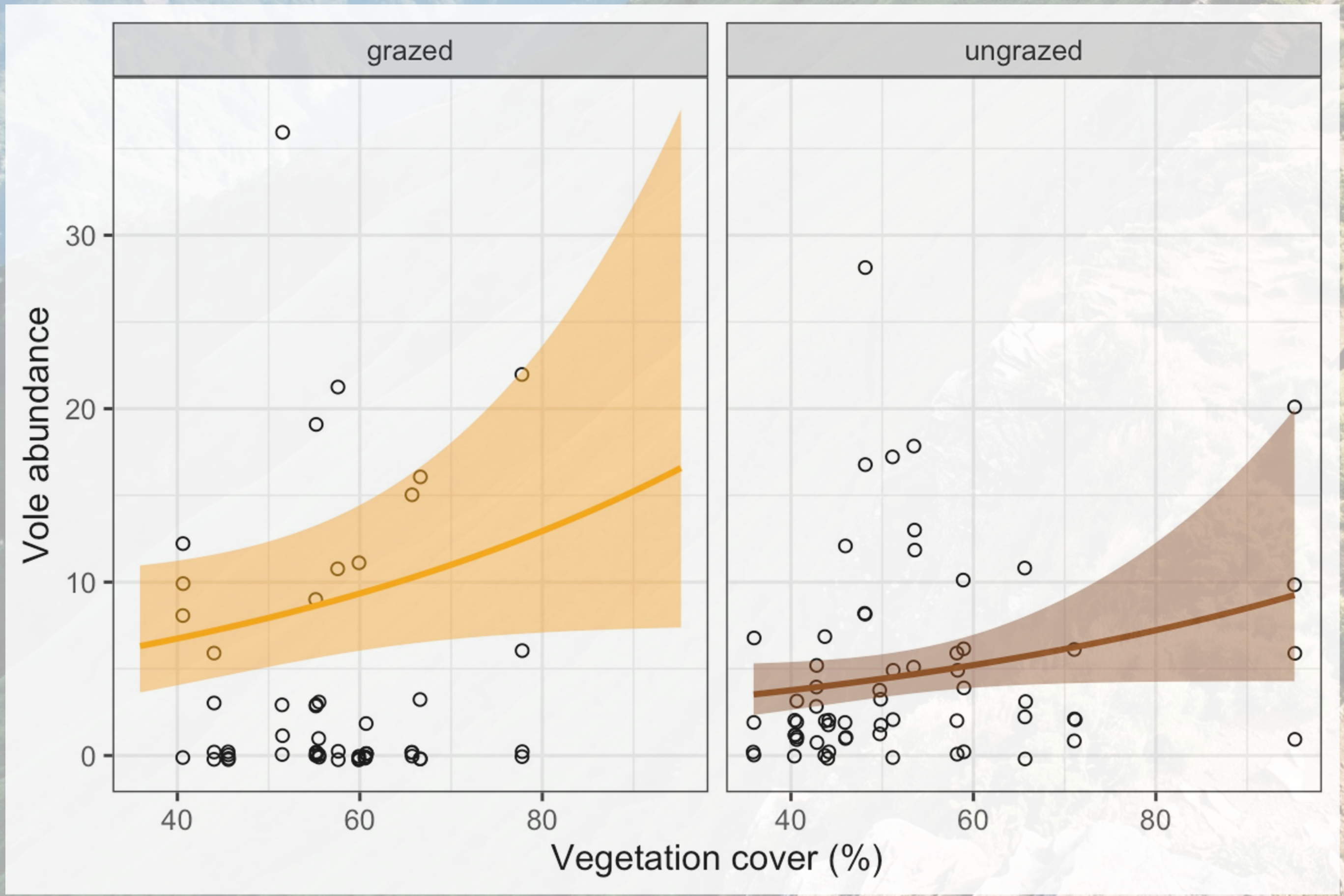
How are excess zeros distributed between grazed and ungraded sites?



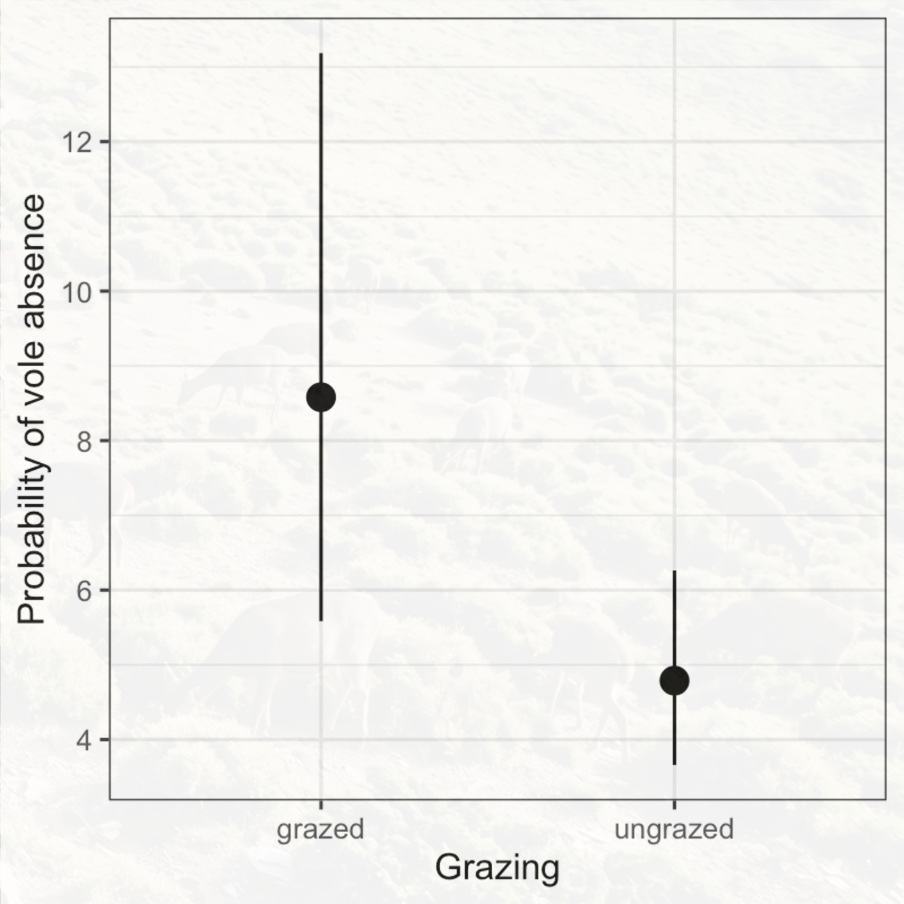
Zero-inflated negative binomial (zinb) model

Two types of responses predicted from a zinb model

count model (non-zero vole acounts as response)



zero-inflation model (binary response)



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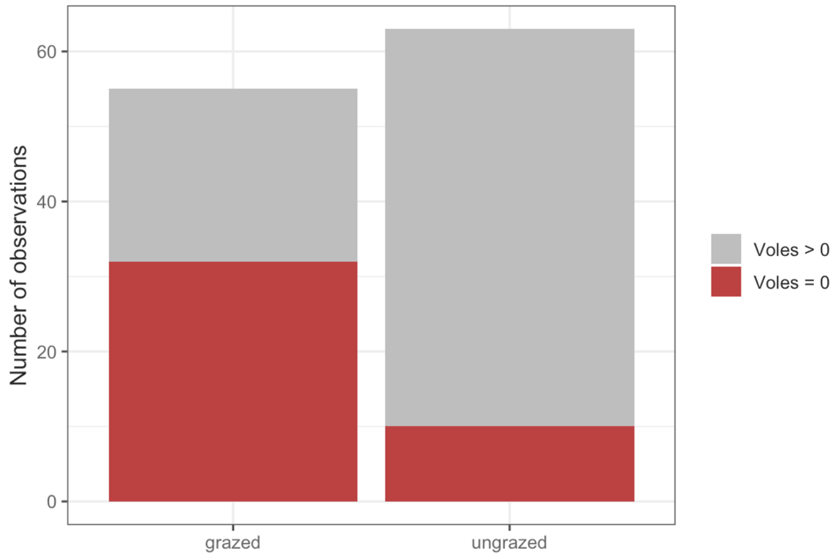
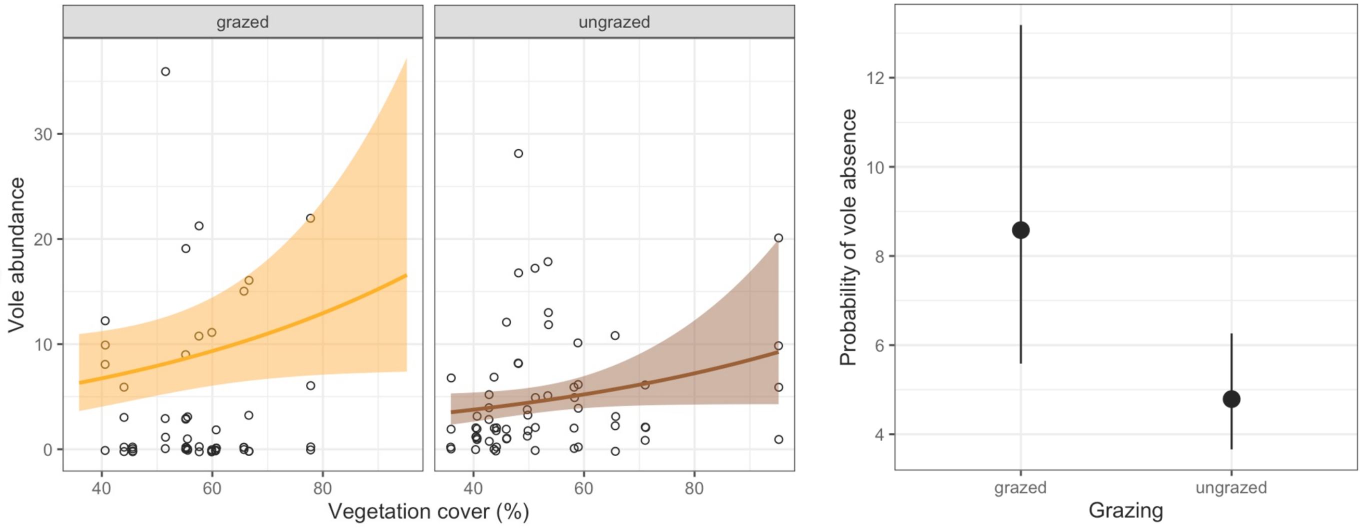
Zero-inflated model or not?

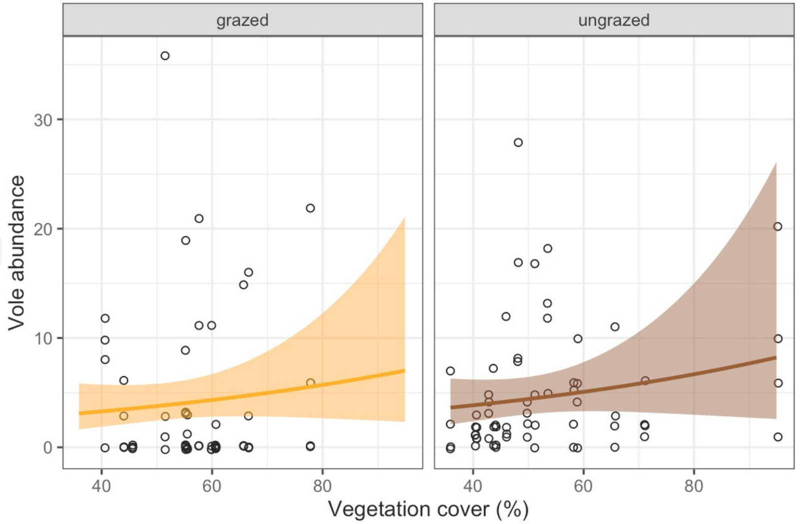
Data exploration analysis for excess zeros



Fit zero-inflated model (e.g., zinb model)

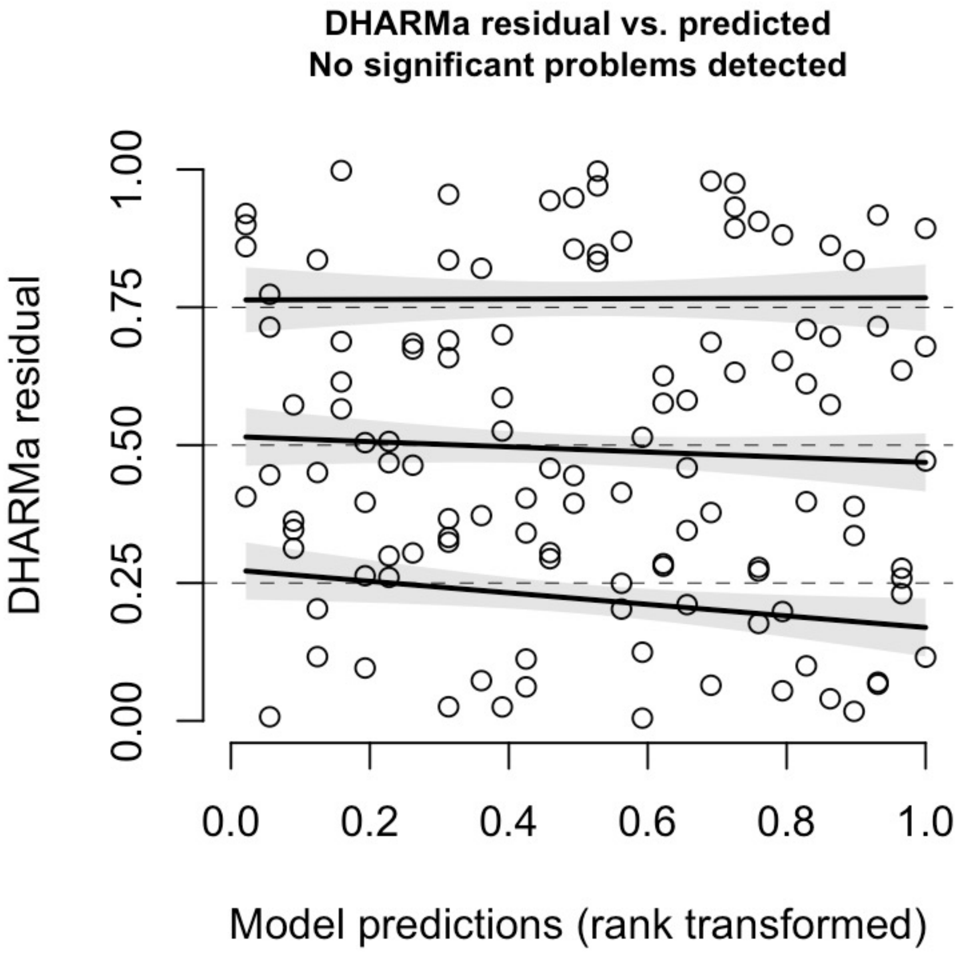


Compare against the model without zero-inflation (e.g., nb model)

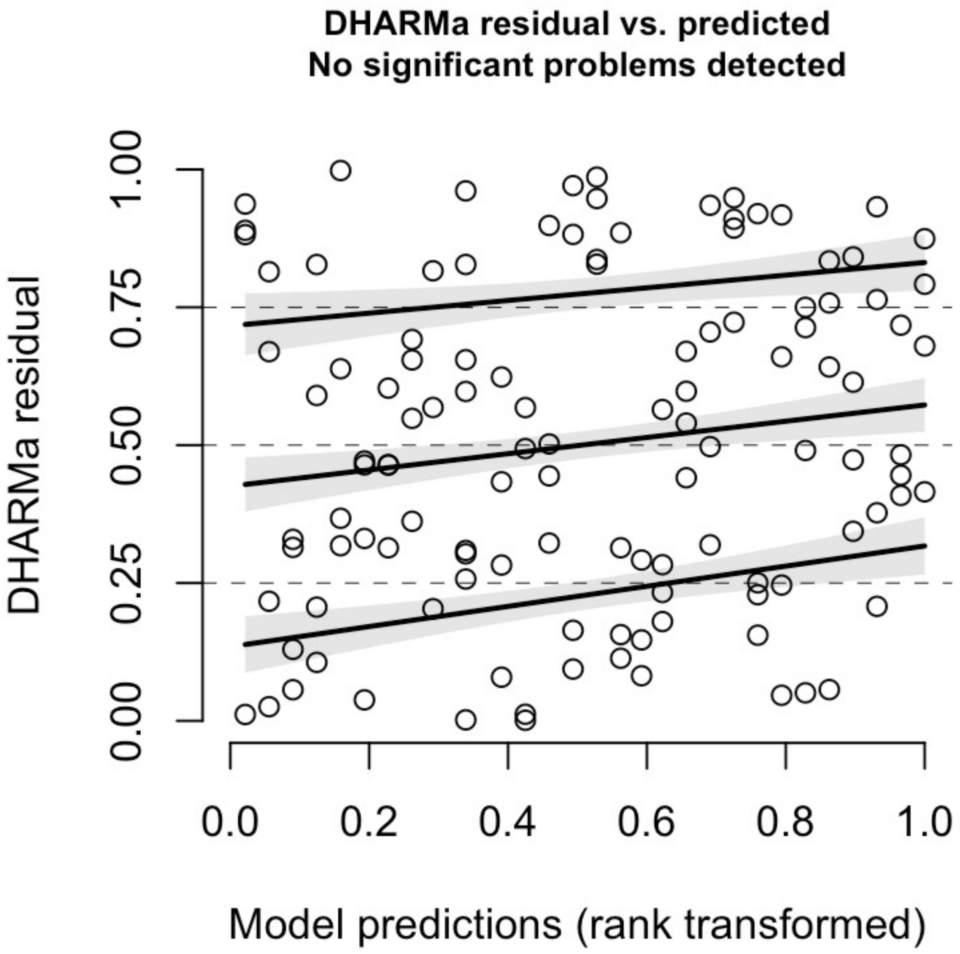


Zero-inflated model or not?

Residuals vs fitted plot from zinb model



Residuals vs fitted plot from nb model

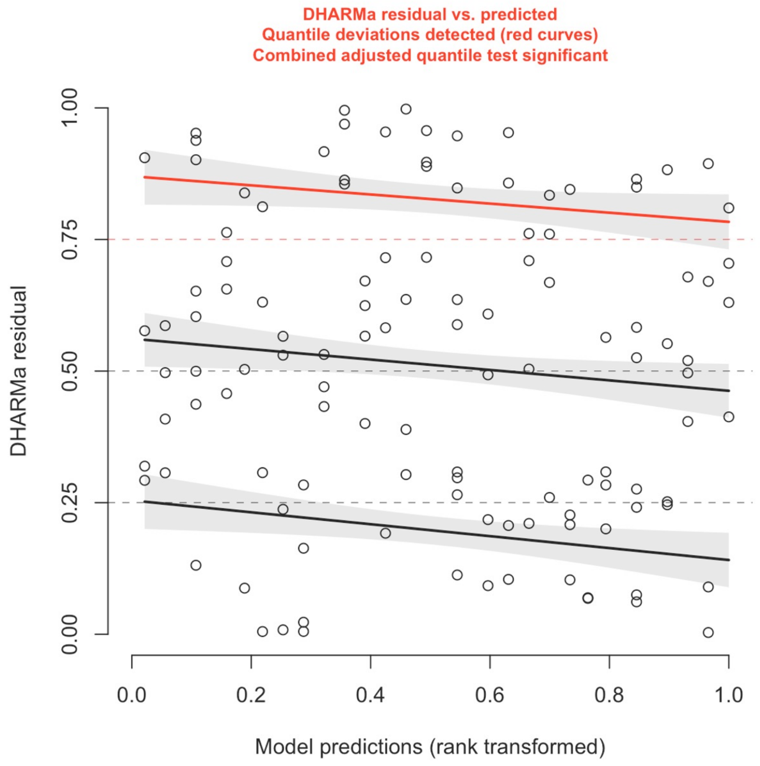


Residuals are sensitive to model fitting

nb model #1 with heteroscedasticity

$voles \sim vegetation\ cover$

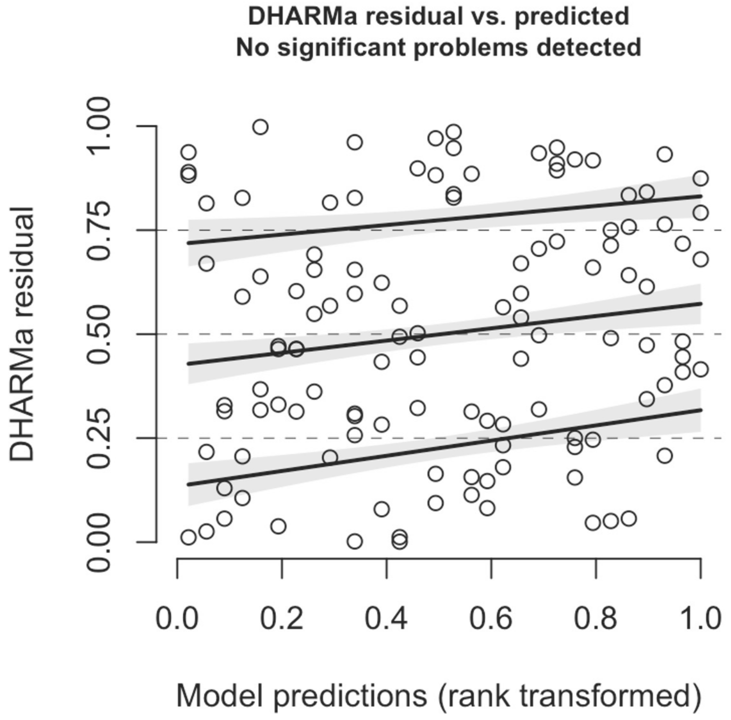
$dispformula \sim vegetation\ cover$



nb model #2 with heteroscedasticity

$voles \sim vegetation\ cover + grazing$

$dispformula \sim vegetation\ cover$

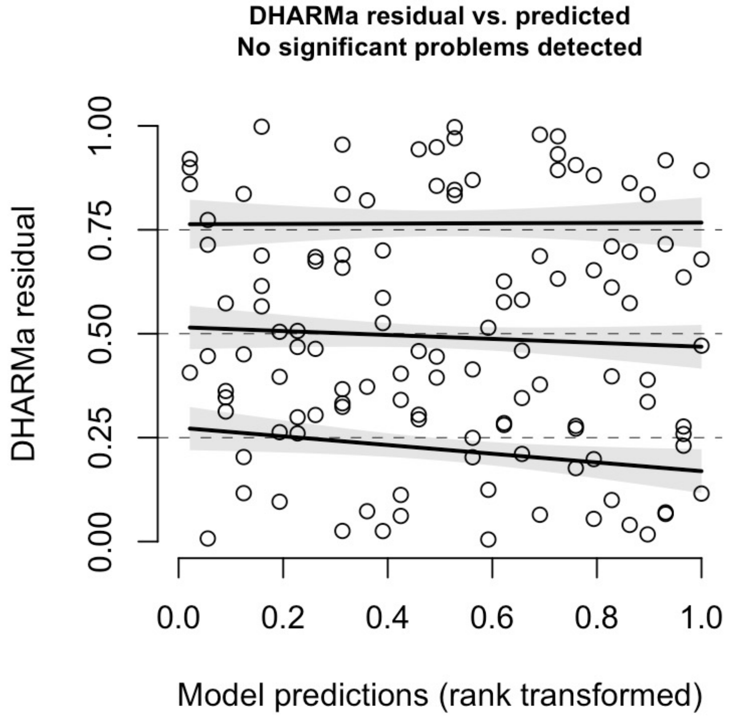


zinb model with heteroscedasticity

$voles \sim vegetation\ cover + grazing$

$dispformula \sim vegetation\ cover$

$zipformula \sim grazing$



Let the biology guide the statistical modeling

- *A complicated model is not more accurate than a simple model*
- *A p value < 0.0001 is not prettier than a p value = 0.2*